NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-39012/34A
11 January 1973
SUPERSEDING
MIL-C-39012/34
27 February 1967

PERFORMANCE SPECIFICATION

CONNECTORS, COAXIAL, RADIO FREQUENCY

(SERIES TNC (UNCABLED) - RECEPTACLE, SOCKET,

HERMETIC SEALED, JAM NUT MOUNTED, CLASS 2)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-PRF-39012.

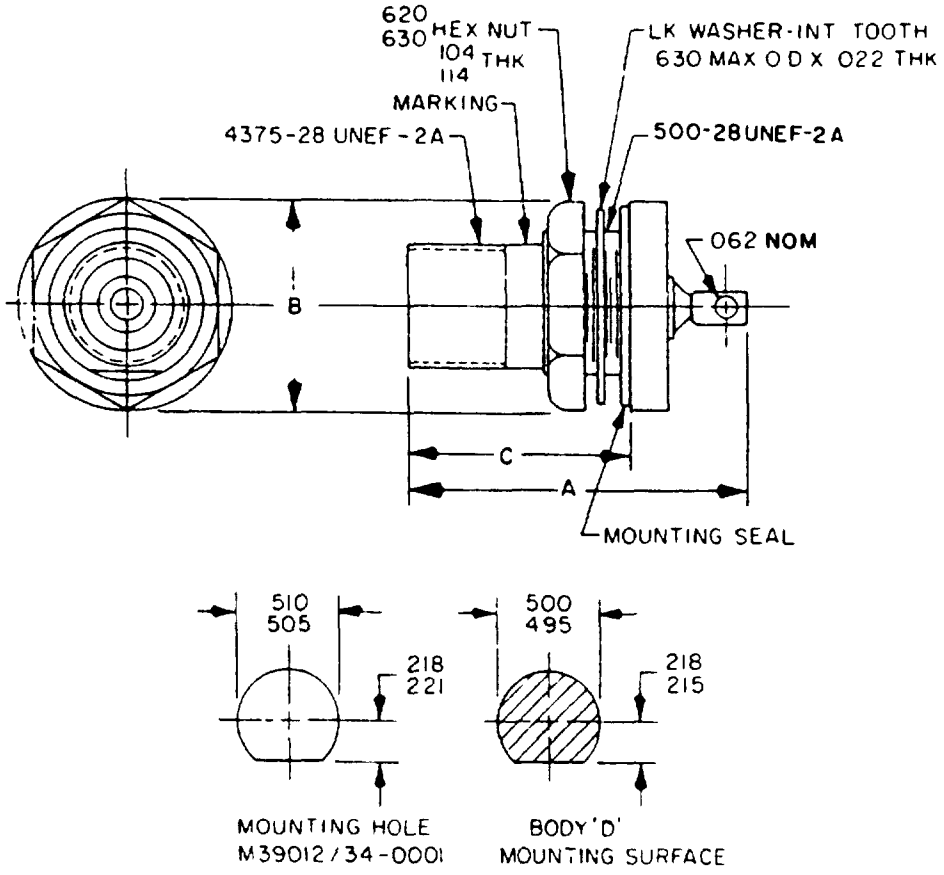
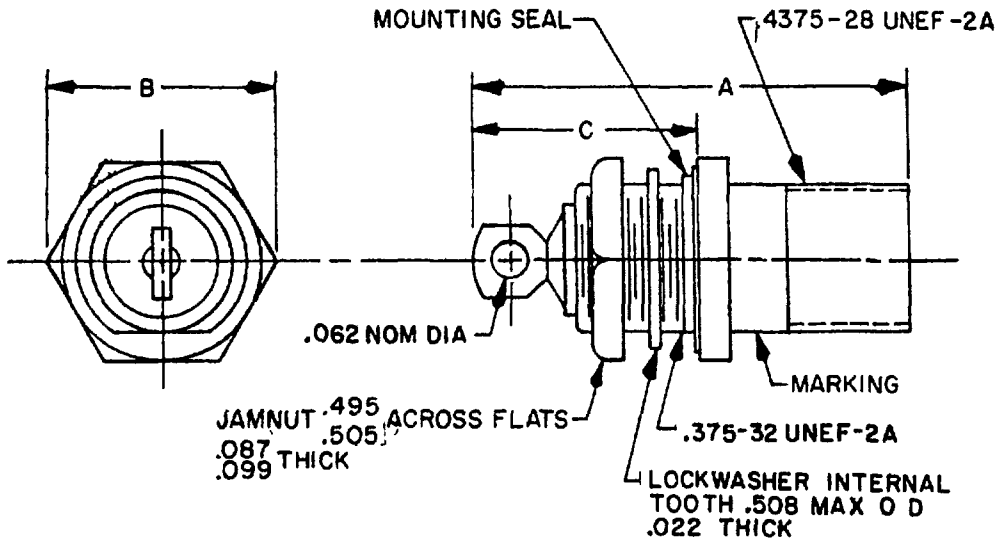
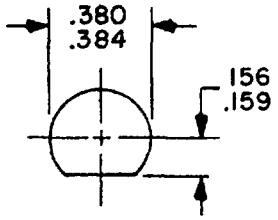


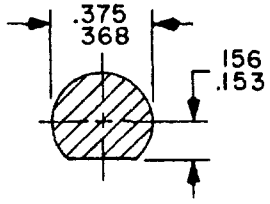
FIGURE 1 General configuration

FSC 5935

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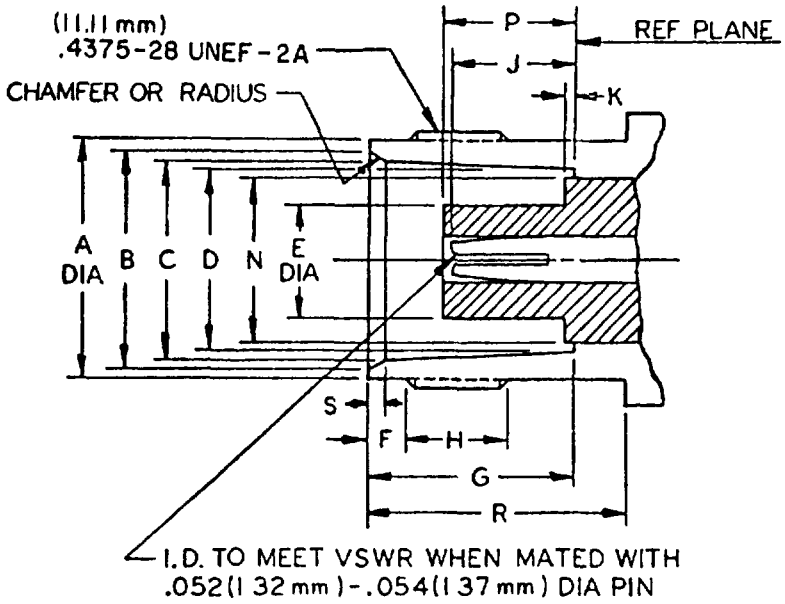


INCHES	MM	INCHES	MM	INCHES	MM
.022	.56	.159	4.04	.4375	11.11
.062	1.57	.215	5.46	.495	12.57
.087	2.21	.218	5.54	.500	12.70
.099	2.51	.221	5 61	505	12.83
.104	2.64	.368	9.35	508	12.90
.114	2.90	.375	9 53	.510	12.95
.153	3.89	.380	9.65	.620	15.75
.156	3.96	.384	9.75	.630	16.00

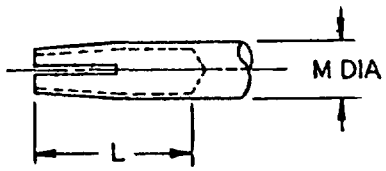
NOTES:
1. Dimensions are in inches.
2. For dimensions A, B, and C see table I.
3. Dimension B is the largest overall diameter of the connector.
4. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
5. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 1. General configuration - Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.378 (9.60)	.381 (9.68)
B	.345 (8.76)	.356 (9.04)
C	.327 (8.31)	.333 (8.46)
D	.319 (8.10)	.321 (8.15)
E		.186 (4.72)
F	.068 (1.73)	.088 (2.24)
G	.329 (8.36)	.333 (8.46)
H	.187 (4.75)	
J	.186 (4.72)	.206 (5.23)
K		.006 (.15)
L	.195 (4.95)	
M	.081 (2.06)	.087 (2.21)
N		.256 (6.50)
P	188 (4.78)	.208 (5.28)
R	.415 (10.56)	
S	.015 (.38)	.030 (.76)



*N dimension applies to that portion (if applicable) of the dielectric which protrudes beyond the metal shoulder (or reference plane) by dimension K.

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 2. Mating dimensions for female terminations.

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TABLE I. Dash number and overall dimensions.

Dash No.	Dim	Inches-millimeters 1/		Maximum panel thickness
		Minimum	Maximum	
0001	A		1.343 (34.11)	
	B	.684 (17.37)	.690 (17.53)	.125 (3.18)
	C	.769 (19.53)	.893 (22.68)	
0002	A		1.328 (33.73)	
	B	.630 (16.00)	.650 (16.51)	.250 (6.35)
	C	.626 (15.90)	.750 (19.05)	

1/ Millimeters are in parentheses.

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
I	0001 0002	0001 0002

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ENGINEERING DATA

Nominal impedance: 50 ohms.

Frequency range 0 to 11,000 MHz.

Voltage rating. 500 volts rms maximum working voltage at sea level, 125 volts rms maximum at 70,000 feet

Temperature rating -65°C to +165°C.

REQUIREMENTS

Dimensions and configuration See figure 1

Force to engage and disengage
Longitudinal force - Not applicable
Torque - 2 inch-pounds maximum

Coupling proof torque Not applicable

Inspection conditions:
Coupling torque - 4 to 6 inch-pounds

Mating characteristics See figure 2 for dimensions.
Center contact (female)
Oversize test pin - .057 diameter minimum (non-closed entry contacts only).
Insertion depth - 125 minimum.
Number of insertions - 1.

Insertion force test - Steel test pin dia .054 minimum.
Test pin finish - 16 microinches.
Insertion force - 2 lbs maximum.

Withdrawal force test Steel test pin dia .052 maximum
Withdrawal force - 2 oz minimum
Test pin finish - 16 microinches

Hermetic seal Leakage shall not exceed 1×10^{-5} cc/sec of tracer gas at atmospheric pressure

Leakage (pressurized connectors) Connector mounted in mounting hole specified on figure 1 with mating end capped Test applicable to mounting seal only Air pressure - 30 psi. Duration. 30 seconds minimum

Insulation resistance Method 302, test condition B, MIL-STD-202. 5,000 megohms minimum

Center contact retention
6 lbs minimum axial force
4 inch-ounces radial torque minimum

Corrosion (salt spray) Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR)
Not applicable.

Connector durability· 500 cycles at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance In milliohms maximum

	Initial	After environment
Center contact	4.0	5.0
Outer contact	.2	Not applicable

Dielectric withstanding voltage Method 301 of MIL-STD-202. 1,500 volts rms minimum at sea level.

Vibration, high frequency Method 204, test condition C, MIL-STD-202.

Shock. Method 213 of MIL-STD-202, test condition I

Temperature cycling· Method 102, test condition C, MIL-STD-202, except test high temperature shall be +200°C.

Thermal shock Not applicable

Moisture resistance Method 106 of MIL-STD-202 No measurements at high humidity Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity

Corona level
Voltage - 375 volts, minimum.
Altitude - 70,000 feet

RF high potential withstanding voltage.
Voltage and frequency 1,000 volts rms at 5 MHz
Leakage current Not applicable

Cable retention force· Not applicable.

Coupling mechanism retention force·
Not applicable

RF leakage Not applicable

Insertion loss Not applicable

Part number M39012/34-
(dash number from table I)

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, MI, EL
Navy - EC
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - AT, AV, ME
Navy - AS, OS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - FS

(Project 5935-1800)

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Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-C-39012/31A CONNECTORS, COAXIAL, RADIO FREQUENCY SERIES TNC (UNCABLED - RECEPTACLE, SOCKET, HERMETIC SEALED, JAM NUT MOUNTED CLASS 2)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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